

A Review of Myopia Control with Atropine

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Abstract

Myopia is a global public health issue with a worldwide prevalence of ~30% and is estimated to rise to 50% by 2050. In addition to the burden associated with routine management of the condition, high myopia predisposes the eye to sight-threatening complications such as myopic maculopathy and glaucoma in adult life. Controlling onset and progression of myopia at a young age can reduce the risk of morbidity associated with high myopia. Progression of myopia can be slowed with various optical, environmental, and pharmaceutical strategies, of which atropine has proven to be the most effective. High-dose atropine (0.5%–1%) is the most effective, but it has significant trade-offs with respect to rebound of myopia on discontinuation and side effects such as photophobia and difficulty with near work (decreased accommodation). Low doses of atropine have been trialed and show a dose-dependent efficacy. However, its mode of action on the ocular tissues leading to slowing eye growth remains unclear and multiple mechanisms and sites in the eye have been postulated to play a role. This review summarizes the role of atropine in controlling myopia and the mechanisms studied to date.

Keywords: myopia, myopia control, atropine, muscarinic antagonists

Background

IN THE UNITED STATES, the prevalence of myopia (near-sightedness) has nearly doubled in just over 30 years from 25% in 1972 to 41.6% in 2004.¹ Even more startling is its prevalence in many East Asian countries such as China, Taiwan, Hong Kong, Singapore, and Korea where ~42.4% to 96.5% of children aged between 6 and 19 years are affected by the condition.^{2–5} Importantly, the prevalence of high myopia (worse than –5.00D) is also high with 2.8% of the world's population being affected in 2010.⁶ Considered to be one of the leading public health issues of our times, evidence shows a rising prevalence of myopia and high myopia, with estimates indicating that myopia will affect between 23% and 66% of the global population by 2050.⁶ That puts an estimated 5 billion people at risk of developing myopia, of which 1 billion (10%) people are at risk of developing high myopia.⁶

Myopia, a progressive condition, is a leading cause of uncorrected refractive error, and the annual global cost relating to the decrease of productivity associated with untreated refractive error was more than US\$200 billion over 5

years.^{7,8} Beside the fact that uncorrected myopia is one of the leading causes of visual impairment,⁹ pathological changes associated with axial elongation such as chorioretinal degeneration or myopic choroidal neovascularization also significantly increase the risk of visual impairment and/or blindness. Indeed, studies from Copenhagen¹⁰ and Rotterdam¹¹ report that myopic degeneration is among the leading causes of visual impairment in the population, while the primary cause of blindness in Japan is myopic macular degeneration.¹² In addition, high myopia also increases the risk of vision impairment and other serious conditions such as glaucoma^{13,14} and retinal detachment.^{15,16} Slowing myopic progression can prevent the risk of complications and the future burden of visual impairment and blindness.

Myopia commonly results from axial elongation of the eye. The annual progression of eye length in children varies between 0.1 and 0.45 mm and is influenced by age, family history, and ethnicity.^{17,18} In terms of spherical equivalent, mean myopia progression in children aged from 6 to 15 years of age is estimated to range between 0.4D/year¹⁹ and 1.0D/year or more^{20–22} for Caucasian and Asian eyes, respectively. The exact mechanism underlying progression of myopia is

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not fully understood, but genetic and environmental factors appear to play a role.²³ However, the rapid pace at which the prevalence of myopia increased over the recent decades is suggestive of a greater role for environmental factors. Moreover, in both animal models and observational and interventional studies involving humans, the eye was found to detect spatial blur and modulate eye growth accordingly.^{24,25} In this respect, it appears that there are local mechanisms across possibly both the central and peripheral retina that play a role, and indeed, recent studies suggest that the peripheral retina plays a significant role with studies demonstrating that the peripheral retina can drive emmetropization in spite of a nonfunctional fovea.²⁶ Other evidence for environmental influence is from data showing that excessive near work or limited outdoor time plays a role in the onset and progression of myopia.^{27,28}

Myopia Control and Atropine

There have been significant efforts in slowing the progression of myopia using optical and pharmaceutical interventions. Despite promising data for myopia control using optical strategies,^{29–33} the effectiveness of these strategies is generally lower than that observed with pharmaceutical interventions (Fig. 1). Overall, atropine, a nonselective antimuscarinic agent, followed by pirenzepine, a selective M1 antimuscarinic agent, shows greater efficacy than orthokeratology or multifocal and peripheral defocus-based optical interventions in slowing progression of myopia.³⁴ Interestingly, although the use of pirenzepine demonstrated an efficacy of up to 50% in slowing myopia progression and also resulted in less pupillary dilation compared with atropine eye drops,^{35,36} it is no longer pursued for further investigations or

commercialization and the reasons remain unclear. On the other hand, atropine has risen in popularity, has been investigated in a number of research studies around the world,^{37–43} and is broadly adopted by many practitioners worldwide to manage myopia.⁴⁴

Higher concentration of atropine (commonly 1%) was initially investigated and found to be substantially effective in slowing axial elongation between 70% and up to 94% in well-conducted trials (Level I and II evidence).^{38–40,45–47} Despite the observation of side effects such as photophobia and difficulty with near work, there were no serious adverse events reported even with long-term follow-up of more than 10 years.^{48–51} Moreover, electroretinograms of eyes treated with atropine showed no evidence of damage in the retina relating to the daily usage of atropine^{52,53} and no difference of intraocular pressure in eyes using atropine eye drops compared with placebo drops.⁵⁴ In addition, atropine demonstrates a dose-dependent efficacy as evidenced by data from a number of studies and the side effects also appear to share dose-dependency.^{34,38,39,42} Even though there were no serious side effects reported to date, photophobia and difficulty with near work remain major concerns especially with higher doses and reported by 38% to up to 100% of participants.^{40,41} It may be that these side effects are the major barriers for initiating the use of atropine in children or for the observed low compliance of atropine in the United States and European countries.⁵⁵ Another concern with the use of atropine is the rebound of myopia that occurs on cessation of atropine eye drops^{56,57} and is particularly an issue with higher doses.

Thus, lower concentrations of atropine that resulted in reduced number of side effects and rebound of myopia were in favor with most practitioners (Table 1),⁵⁸ and interestingly, in a 5-year study atropine for the treatment of

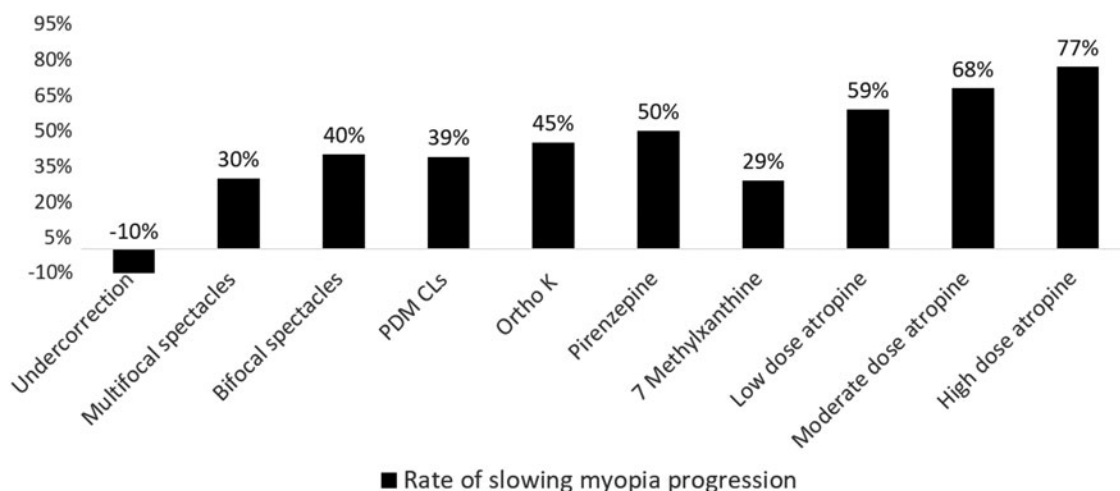


FIG. 1. Comparison of effectiveness between different interventions for myopia control treatment. There are different interventions contributing to the myopia control treatment; the figure above summarizes the effectiveness of each individual treatment strategy in slowing myopia progression. The vertical axis, the y-axis, represents the effectiveness that relates to the difference between the mean annual ocular elongation of the treatment group and the control group. The horizontal axis, the x-axis, represents different interventions for myopia control as follows: (1) Undercorrection (Adler et al. 2006⁵⁹); (2) Multifocal spectacles (Sankaridurg et al.³¹); (3) Bifocal spectacles (Cheng et al. 2014⁶⁰); (4) PDM CLs: peripheral defocus-modifying contact lenses (Sankaridurg et al.³³); (5) Ortho-K: orthokeratology (Cho et al.³⁰); (6) Pirenzepine: selective muscarinic receptor antagonists (Tan et al.³⁶); (7) 7 methylxanthine: adenosine receptor antagonists (Trier et al. 2008⁶¹); (8) Low-dose atropine: 0.01% atropine, nonselective muscarinic receptor antagonist (Chia et al.³⁹); (9) Moderate-dose atropine: 0.1% atropine (Chia et al.³⁹); and (10) High-dose atropine: 0.5% atropine (Chia et al.³⁹).

TABLE 1. SUMMARY OF EFFECTIVENESS OF DIFFERENT CONCENTRATIONS OF ATROPINE EYE DROPS

Concentration of atropine	Study	Annual change in spherical equivalent (D)
High (0.5%–1%)	Brodstein 1984—1%	0.55D (88.7%)
	Yen 1989—1%	0.69D (75.8%)
	Shih 2001—0.5%	0.65D (69.9%)
	Kennedy 2000—1%	0.31D (86.1%)
	Synuita 2001—1%	0.79D (94.0%)
	Chua 2006—1%	0.92D (76.7%)
	Polling 2016—0.5%	0.40D (80.0%)
	Wang 2017—0.5%	1.20D (60.0%)
Moderate (0.1%–0.49%)	Lee 2016—0.125%	1.00D (95.2%)
	Chia 2012—0.1%	0.82D (68.3%)
Low (<0.1%)	Lee 2006—0.05%	0.47D (62.7%)
	Clark 2015—0.01%	0.50D (83.3%)
	Chia 2012—0.01%	0.71D (59.1%)

myopia 2 (ATOM2), 0.01% atropine was found to be more effective than higher doses in slowing myopia progression with reduced side effects.⁵⁸

Atropine: How Does it Slow Myopia Progression?

Atropine sulfate, widely used as eye drops, is the sulfate salt of atropine, an alkaloid that can not only be derived from the leaves of *Atropa belladonna* but is also found in other plants mainly from the Solanaceae family such as *Datura stramonium*, *A. belladonna*, *Hyoscyamus niger*, and *Mandragora officinarum*. Atropine consists of an organic base (tropine) and an aromatic (tropic) acid to complete the structure of an organic ester. Summarizing from the basics in pharmacology, atropine is a nonselective, antimuscarinic receptor agent with an affinity for all 5 subtypes of muscarinic acetylcholine M1 to M5 receptors. The tropine base and tropic acid itself do not possess antimuscarinic activity and the ester is considered to be responsible for the activity.⁶² In the eye, atropine was used topically at up to 4% of concentration.⁶² Two main effects of topical application of 1% atropine (the most commonly used concentration) are mydriasis (dilation of pupil) and cycloplegia (paralysis of ciliary muscle, resulting in loss of accommodation). While the mydriatic effect commences 30 min after being instilled with full recovery from the effect in 7 to 10 days, the cycloplegic effect commences 40 min following instillation with full recovery in 10 days to 2 weeks. In clinical settings, other than its extensive application for cycloplegic refraction, atropine eye drops have been used for amblyopia treatment,⁶³ spasm of near reflex,^{64,65} and myopia control.

Various mechanisms were postulated over the years to explain the role of atropine in slowing eye growth in myopic eyes, but as yet, its mechanism of action remains unclear. Earlier studies considered that the compound exerted its efficacy by its cycloplegic action on the smooth ciliary muscle and blocking the accommodative function of the eye.^{24,66} However, studies in animal models demonstrated that sectioning of the optic nerve⁶⁷ or lesioning of the Edinger–Westphal nucleus⁶⁸ did not inhibit development or recovery of experimental myopia and therefore indicated that atropine

might be exerting its action through routes other than accommodative mechanisms. In addition, atropine was found to inhibit myopia development in chicks wherein no difference in carbachol-induced accommodation or pupillary constriction to light was observed between atropine- and saline-injected chicks and thus indicated that atropine may be acting through a nonaccommodative mechanism.⁶⁹

Thereafter, it was suggested that atropine may be exerting its action by altering retinal neurotransmission. The amacrine cells of the retina have muscarinic receptors, but ablation of the cholinergic amacrine cells did not prevent atropine from inhibiting axial elongation,⁷⁰ and it was concluded that atropine may be exerting its growth-suppressing influence by acting on extraretinal muscarinic receptors, possibly in the retinal pigment epithelium, choroid, or sclera.

The choroid is a vascular structure that plays an active role in emmetropization by changing thickness and moving the retinal image plane in response to optical defocus. Muscarinic antagonists, including atropine, were found to cause rapid, transient choroidal thickening, and it was suggested that the thickening response of the choroid and ocular growth inhibition may be mechanistically linked.⁷¹ The mechanism leading to increased choroidal thickness is not clear as in other independent experiments, the use of atropine did not result in alteration of choroidal blood flow in response to light/dark transitions⁷² or during isometric exercises.⁷³

Interestingly, muscarinic receptors are also found in the retinal pigment epithelium and therefore could be a potential site of action. The retinal pigment epithelium, with respect to muscarinic receptors, is suspected to be a relay, transferring the signaling cascade toward the target tissue—the choroid or the sclera.^{74,75} Atropine was found to increase the release of dopamine, but reduced the electroretinogram (ERG) *b* and *d* waves and damped oscillations of retinal pigment epithelium (RPE) potentials. It was suggested that by dampening vital functions of the retina, atropine boosts dopamine release from cellular stores, which then controls eye growth.⁷⁶ In many experimental animal studies, the use of either dopamine or nonselective dopamine receptor agonists was found to inhibit development of myopia. Clearly, further work is needed to understand if atropine is exerting its action through the retinal pigment epithelium and acting on dopamine pathways.

Additionally, cultured human scleral fibroblasts demonstrate mRNA expression of the 5 muscarinic receptors, indicating that sclera may be a potential site of action.⁷⁷ In support of the role of atropine in affecting sclera, chick scleral chondrocytes were said to contain muscarinic acetylcholine receptors and application of atropine resulted in inhibition of the synthesis of DNA and glycosaminoglycans from scleral chondrocytes.⁷⁴ In addition, real-time PCR showed upregulation of mRNA levels of M1, M3, and M4 receptors in the sclera after subconjunctival atropine treatment in addition to lens-induced myopia in mice, compared with no changes found in the group of mice having lens-induced myopia without atropine and the normal group of mice.⁷⁸

Challenges and Gaps

Although evidence for effectiveness of atropine in slowing eye growth is consistent, the mechanism of action remains to be elucidated. Additionally, with respect to low-dose atropine, the observed lack of effect on axial length needs to be

explored further. Furthermore, while bifocal or multifocal spectacle lenses are commonly prescribed to minimize the near vision disturbances associated with the use of atropine, such spectacle lenses were independently found to have a role in slowing progression of myopia. While the efficacy of atropine in slowing eye growth shows irrefutable proof from its use in various animal models, in humans, the benefit, if any, accrued from the use of such bifocals or multifocals has not been separated. In addition, although there have been no serious side effects of long-term use of atropine, it is not still known whether long-term mydriasis or cycloplegia may increase macular degeneration or cataract.

Conclusion

In summary, the effect of atropine eye drops in controlling myopia progression is backed by substantial clinical evidence. Although there are unanswered questions around the efficacy of low doses of atropine, there is no doubt about the efficacy of the compound in slowing eye growth. A bigger challenge is the lack of understanding of the site of action and pathways driving the reduced axial elongation. A better understanding of the site of action would help exploit and apply technologies that could target the site to maximize efficacy while minimizing effect on other tissues and therefore reducing side effects.

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